Estimators and the method of moments

This note is a short bridge for outcome S7. It standardises the language used in Week 7; it does not replace the assigned point-estimation and Penn State readings. The examples below are original institutional examples.

1. Sampling notation and the objects being estimated

For this note, let $n \geq 2$, let θ be a scalar parameter in a stated parameter space $\Theta \subseteq \mathbb{R}$, and let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} F_\theta.$$

The observations are random variables before sampling and their realised values are $x_1, \ldots, x_n$. We use the course conventions

$$\bar{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad S^2 := \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

with observed values $\bar{x}$ and s^2. The denominator $n-1$ in S^2 is part of this convention; it is not the denominator used for an n^{-1} raw or central sample moment in the method of moments.

The following distinctions prevent most terminology errors.

Object	Meaning before the data are observed	Typical notation
Population parameter	A fixed, unknown value indexing the population model.	$\theta \in \Theta$
Estimand	The population quantity to be learned; it may be $\tau(\theta)$, not only θ itself.	$\tau(\theta)$
Statistic	Any function of the sample, hence a random variable.	$T = T(X_1, \ldots, X_n)$
Estimator	A statistic selected to estimate the estimand. For the direct target $\tau(\theta) = \theta$, write $\hat{\theta} = T(X_1, \ldots, X_n)$.	$\hat{\tau} := T(X_1, \ldots, X_n)$
Observed estimate	The number obtained after substituting the realised data.	$\hat{\tau}(\mathbf{x}) = T(x_1, \ldots, x_n)$
Sampling distribution	The probability distribution of an estimator under repeated samples from F_θ, with θ held fixed.	distribution of T under F_θ
Standard error	The standard deviation of the estimator's sampling distribution.	$\mathrm{se}_\theta(\hat{\tau}) := \sqrt{\mathrm{Var}_\theta(\hat{\tau})}$

Thus $\hat{\theta}$ is random, whereas $\hat{\theta}(\mathbf{x})$ is an observed number. A statistic need not be an estimator: for example, $\max_i X_i$ is a statistic, but it is an estimator only if a stated estimand and purpose give it that role.

For a population mean with finite variance $\mathrm{Var}_\theta(X_i) = \sigma^2$, $\mathrm{se}_\theta(\bar{X}) = \sigma/\sqrt{n}$. If σ is unknown, the usual plug-in standard-error estimate is $S/\sqrt{n}$. This is sampling variability, not the estimator's bias and not its mean squared error.

2. Bias, variance and mean squared error

Write the target as $\tau = \tau(\theta)$. Provided the required expectations exist, the bias and variance of an estimator $\hat{\tau}$ are

$$\mathrm{Bias}_\theta(\hat{\tau}) := \mathbb{E}_\theta[\hat{\tau}] - \tau, \qquad \mathrm{Var}_\theta(\hat{\tau}) := \mathbb{E}_\theta\big[(\hat{\tau} - \mathbb{E}_\theta[\hat{\tau}])^2\big].$$

The estimator is **unbiased** for τ when its bias is zero for every admissible θ. The mean squared error (MSE) measures squared error around the target:

$$\mathrm{MSE}_\theta(\hat{\tau}) := \mathbb{E}_\theta[(\hat{\tau} - \tau)^2].$$

If $\mathbb{E}_\theta[\hat{\tau}^2] < \infty$, the bias–variance decomposition is

$$\mathrm{MSE}_\theta(\hat{\tau}) = \mathrm{Var}_\theta(\hat{\tau}) + \mathrm{Bias}_\theta(\hat{\tau})^2.$$

To see why, insert and subtract $\mathbb{E}_\theta[\hat{\tau}]$:

$$\hat{\tau} - \tau = (\hat{\tau} - \mathbb{E}_\theta[\hat{\tau}]) + (\mathbb{E}_\theta[\hat{\tau}] - \tau).$$

After squaring and taking expectations, the cross term is zero because the first bracket has expectation zero. The finite-second-moment condition is what makes the displayed variance and MSE finite. A smaller variance does not automatically mean a smaller MSE: a biased estimator can trade bias for less sampling variation.

The standard error is the square root of the variance term only. It is not the square root of MSE unless the estimator is unbiased.

3. Consistency in probability

An estimator sequence $\hat{\tau}_n$ is **consistent in probability** for $\tau(\theta)$ when, for every $\varepsilon > 0$,

$$\mathbb{P}_\theta(|\hat{\tau}_n - \tau(\theta)| > \varepsilon) \longrightarrow 0 \qquad \text{as } n \longrightarrow \infty.$$

We write this as $\hat{\tau}_n \xrightarrow{p} \tau(\theta)$. It says that large samples make a fixed-sized error unlikely; it is not a statement that every estimate is close, nor that the estimator is unbiased at a finite n.

An elementary sufficient check is

$$\mathrm{MSE}_\theta(\hat{\tau}_n) \longrightarrow 0 \quad \Longrightarrow \quad \hat{\tau}_n \xrightarrow{p} \tau(\theta),$$

because Markov's inequality gives

$$\mathbb{P}_\theta(|\hat{\tau}_n - \tau| > \varepsilon) \le \frac{\mathrm{MSE}_\theta(\hat{\tau}_n)}{\varepsilon^2}.$$

For an iid sample with finite variance σ^2, $\mathrm{Var}_\theta(\bar{X}) = \sigma^2/n$, so Chebyshev's inequality gives consistency of $\bar{X}$ for the population mean. By contrast, an unbiased statistic that keeps using only X_1 has variance that does not shrink with n, and need not be consistent. No advanced asymptotic theory is needed for these checks.

4. The one-parameter method of moments

The method of moments (MOM) matches population raw moments with sample raw moments. For a one-parameter model, use the following recipe.

Method

1. **State the model and domain.** Write the support and the parameter space Θ. Confirm that the selected population moment is finite on Θ.
2. **Choose an identifying moment.** For a raw moment of order r, define

$$m_r(\theta) := \mathbb{E}_\theta[X^r], \qquad M_r := \frac{1}{n}\sum_{i=1}^{n} X_i^r.$$

   Usually start with $r = 1$, so $M_1 = \bar{X}$. The map $\theta \mapsto m_r(\theta)$ should identify the parameter, at least on the relevant part of Θ.
3. **Match and solve.** Set

$$m_r(\hat{\theta}_{\text{MOM}}) = M_r$$

   and solve for all candidate values of $\hat{\theta}_{\text{MOM}}$.
4. **Check the candidates.** State whether each moment-matching candidate is in Θ, and check that the observed data lie in the fitted model's support. A total estimating statistic may take a realised value outside Θ; report it as an out-of-space or boundary-valued result, not as an admissible parameter value. A moment equation can also have no solution or several roots when the moment map is not one-to-one. Report the failure or ambiguity; do not silently choose a root or project it to a boundary.
5. **Evaluate and, if requested, assess.** Substitute the observed data to obtain the estimate. Use expectation, variance, MSE or a law of large numbers argument only after checking their moment conditions.

The one-parameter recipe does not require a second equation. With two or more parameters, additional identifying moments would be needed; that is a separate extension of the same idea, not a reason to use an unverified moment equation.

Example: a Poisson mean

Let

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda), \qquad \lambda \in \Theta = (0, \infty),$$

where the support is $\{0, 1, 2, \ldots\}$ and $\mathbb{E}_\lambda[X] = \lambda$. The first-moment equation is therefore

$$\hat{\lambda}_{\text{MOM}} = \bar{X}.$$

For the observed count data $(x_1, x_2, x_3, x_4) = (2, 0, 1, 3)$,

$$\bar{x} = \frac{2+0+1+3}{4} = \frac{3}{2}, \qquad \hat{\lambda}_{\text{MOM}}(\mathbf{x}) = \frac{3}{2}.$$

The estimate is admissible and the data respect the Poisson support. Under the model,

$$\mathbb{E}_\lambda[\hat{\lambda}_{\text{MOM}}] = \lambda, \qquad \text{Var}_\lambda(\hat{\lambda}_{\text{MOM}}) = \frac{\lambda}{n}, \qquad \text{MSE}_\lambda(\hat{\lambda}_{\text{MOM}}) = \frac{\lambda}{n}.$$

The Poisson second moment is finite, so Chebyshev's inequality gives, for every $\varepsilon > 0$,

$$\mathbb{P}_\lambda(|\bar{X} - \lambda| > \varepsilon) \leq \frac{\lambda}{n\varepsilon^2} \longrightarrow 0.$$

Hence $\hat{\lambda}_{\text{MOM}}$ is unbiased and consistent. If all observed counts are zero, the total estimating statistic still has the realised value $\bar{x} = 0$. This boundary-valued MOM estimate lies outside the specified parameter space $(0, \infty)$, so it

is not an admissible parameter value under this model. This does not make $\bar{X}$ undefined, and the unconditional expectation, variance, MSE and consistency statements above remain valid. If a different model explicitly allowed $\lambda = 0$, the same realised value would be admissible under that model. A negative observed count is outside the stated support and cannot be treated as an ordinary Poisson sample.

5. Candidate checks and the elementary likelihood contrast

Moment matching is not a guarantee of a valid estimator. Before reporting a MOM answer, check all of the following:

- **Domain:** is the candidate in the declared Θ, including any endpoint convention?
- **Support:** are the realised data possible under the fitted distribution?
- **Identification:** does the chosen moment map have one admissible root?
- **Moment existence:** are the expectations and variances used in the calculation finite?
- **Interpretation:** have you reported both the random estimator and the observed estimate when both are requested?

The same candidate discipline applies to elementary maximum likelihood. For fixed data $\mathbf{x}$, maximise

$$L(\theta; \mathbf{x}), \qquad \ell(\theta; \mathbf{x}) := \log L(\theta; \mathbf{x}),$$

over the stated parameter space. A score equation gives only an interior candidate. Check curvature or monotonicity, the data support, and every allowed boundary; an open boundary may give a supremum but no attained MLE. This note does not develop advanced likelihood theory.

6. Pitfalls and final checks

- Do not call $\hat{\theta}(\mathbf{x})$ an estimator: it is the observed estimate; $\hat{\theta}$ is random before observation.
- Do not call the standard error a bias or an MSE. Standard error is $\sqrt{\text{Var}}$; MSE also includes squared bias.
- Keep $S^2 = (n-1)^{-1} \sum_i (X_i - \bar{X})^2$ distinct from the n^{-1} raw or central moments used in MOM.
- State finite-moment assumptions before using a variance, MSE or Chebyshev calculation.
- Unbiasedness at each finite n does not by itself imply consistency.
- A moment equation may produce no admissible root or more than one root; report the issue and the parameter-space reason.
- A score root, like a moment root, is a candidate until the objective, support and boundaries have been checked.

For the assigned external route, use Pishro-Nik, Section 8.1.1 definitions and Sections 8.2.0–8.2.3 and Penn State STAT 415, primary course author Laura Simon, Lesson 3 Section 3.2 only at the locators specified in the Week 7 reading guide. Those resources are **EXTERNAL-LINK-ONLY**: this bridge uses original prose and examples and does not reproduce, adapt or extend their examples. Skip the sections and examples listed in that guide, including advanced MLE material, regression and software-based work.